

Lecture 8 -- Overfitting

Prof. Xiaowei Huang

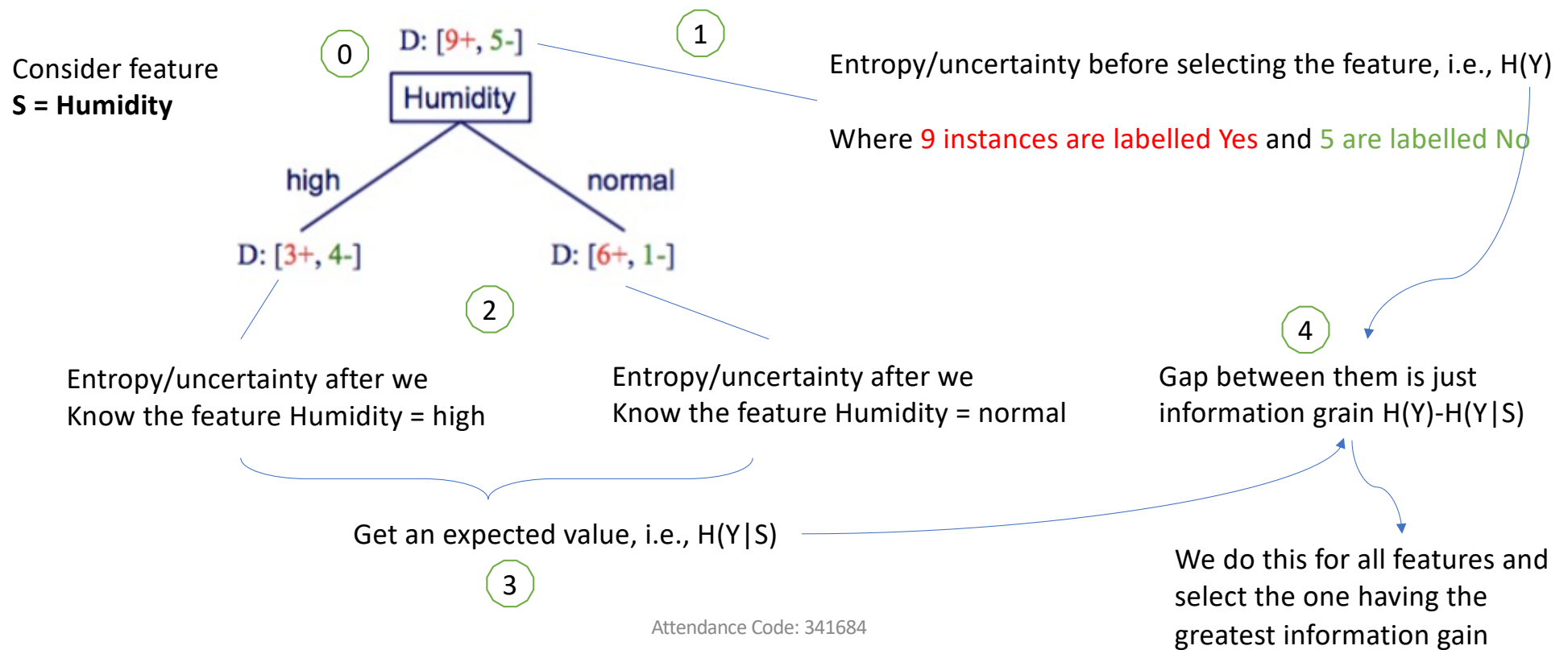
<https://cgi.csc.liv.ac.uk/~xiaowei/>

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Intuition on information gain



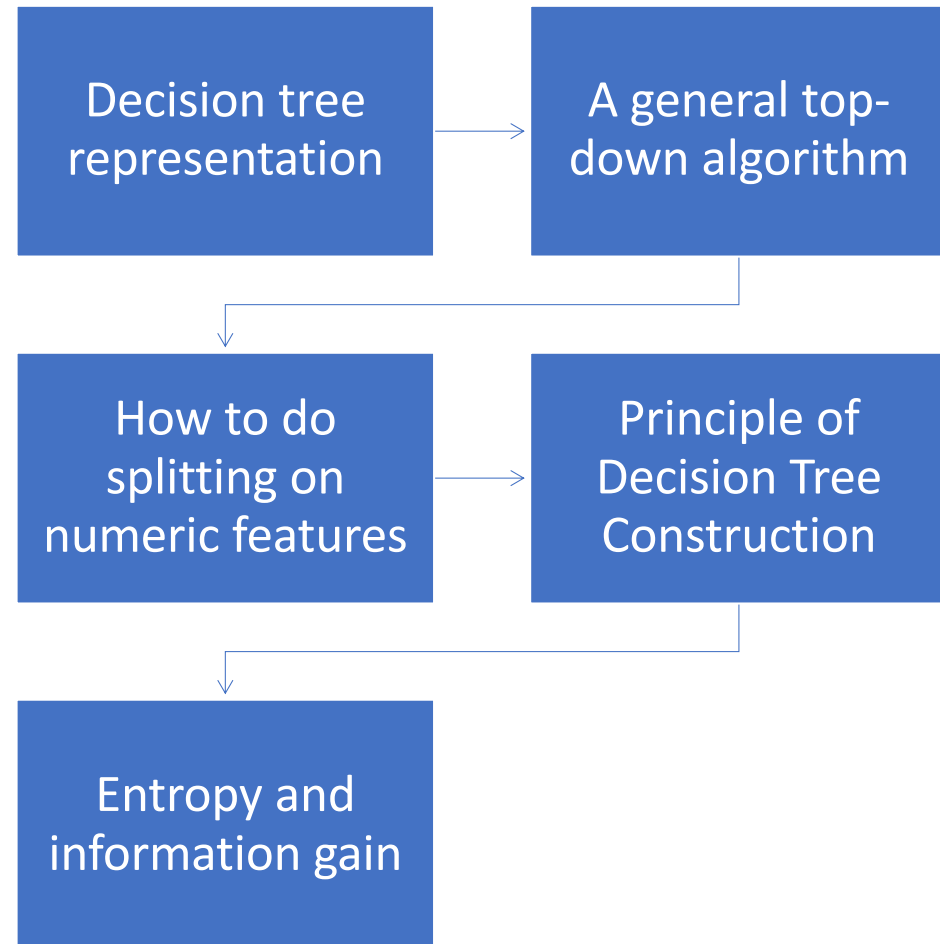
Entropy

- is a key concept that will be used in many places of ML

- E.g., cross-entropy loss $H(P, Q) = - \sum_x P(x) \log Q(x) = -\mathbb{E}_P \log Q$

- E.g., KL-divergence $KL(P|Q) = - \sum_x P(x) \log \frac{P(x)}{Q(x)} = -\mathbb{E}_P \log \frac{P}{Q}$

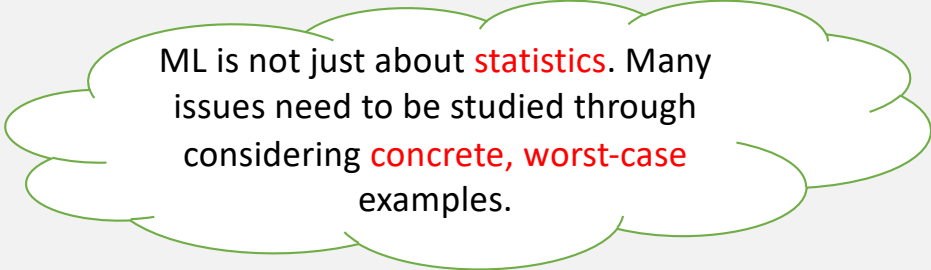
Decision Tree up to now,



Today's Topics

-
1. Stopping criteria of decision trees
 2. Accuracy of decision trees
 3. Pruning and validation dataset

4. Overfitting
5. Overfitting in decision tree



ML is not just about **statistics**. Many issues need to be studied through considering **concrete, worst-case** examples.

Why we care about overfitting?

- (1) A typical, and often discussed, issue in ML.
- (2) An example showing how we analyse a ML issue.



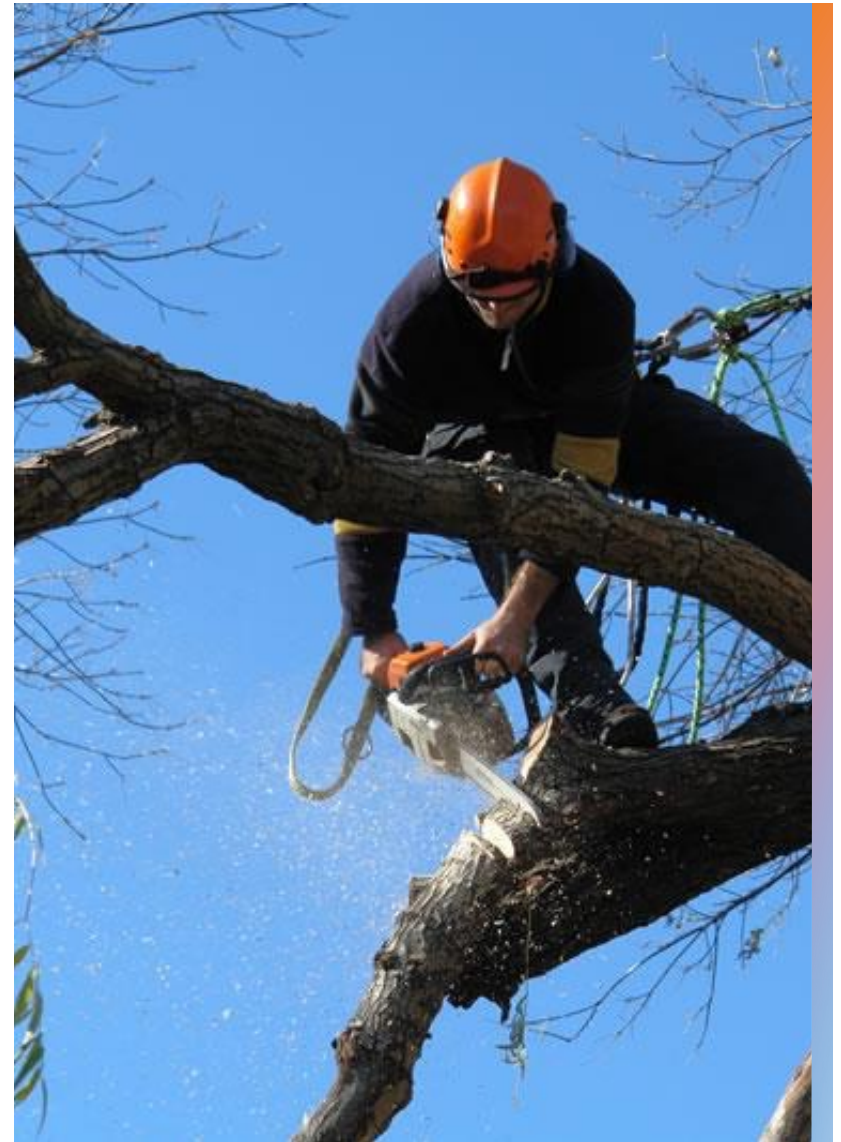
1. Step (3): Stopping criteria

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Stopping criteria

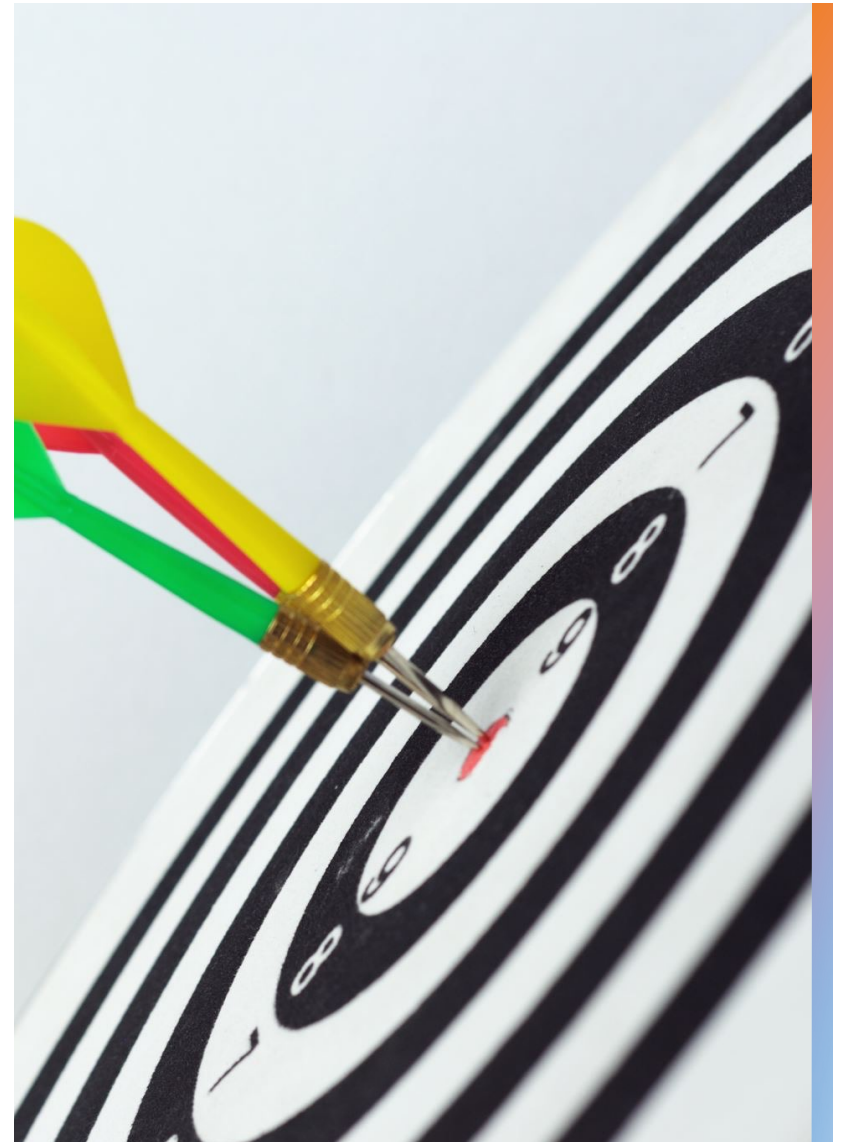
- We should form a leaf when
 - all of the given subset of instances are of the same class, or
 - we've exhausted all of the candidate splits

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2. Accuracy of Decision Tree

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Definition of Accuracy and Error

- Given a set D of samples and a trained model M , the accuracy is the percentage of correctly labeled samples. That is,

$$Accuracy(D, M) = \frac{|\{M(x) = l_x \mid x \in D\}|}{|D|}$$

Where l_x is the true label of sample x and $M(x)$ gives the predicted label of x by M

- Error is a dual concept of accuracy.

But, what is D ?

$$Error(D, M) = 1 - Accuracy(D, M)$$

How can we
assess the
accuracy of a
tree?

D = training dataset

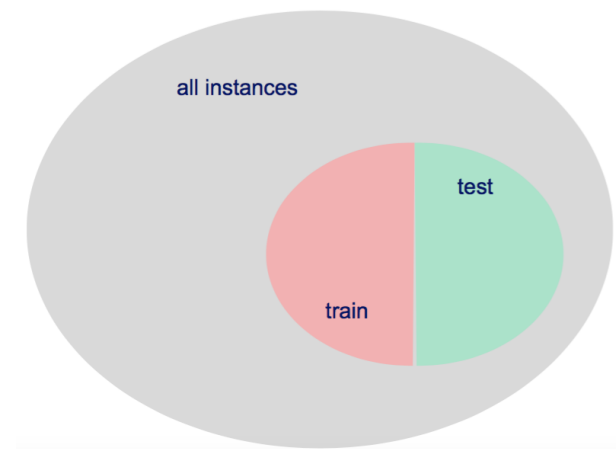
- Can we just calculate **the ratio of training instances** that are correctly classified?

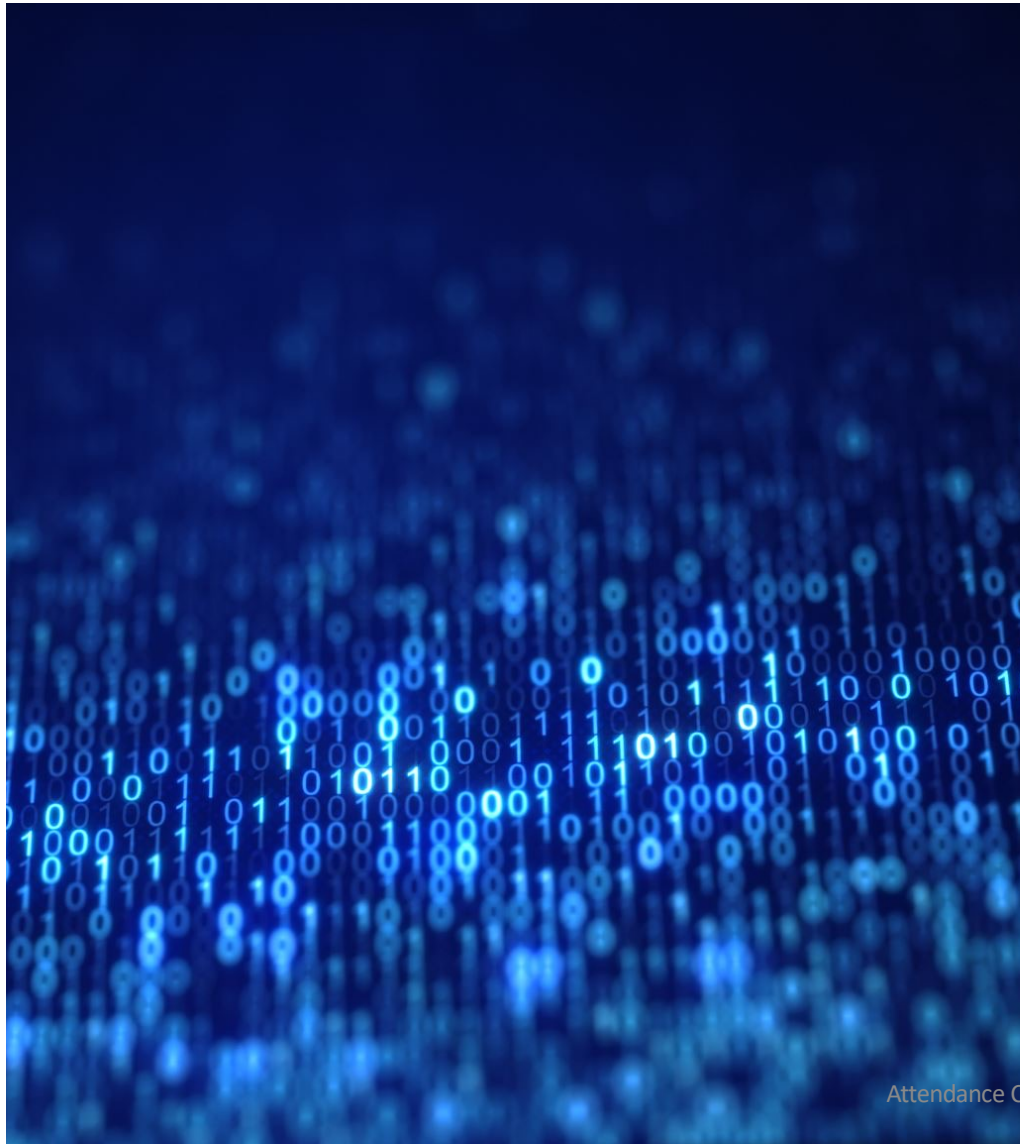
Ground truth

- Consider a problem domain in which instances are assigned labels at random with $P(Y = t) = 0.5$
 - how accurate would a learned decision tree be on previously unseen instances?
 - Can never reach 1.0.
 - how accurate would it be on its training set?
 - Can be arbitrarily close to, or reach, 1.0 if model can be very large.

How can we assess the accuracy of a tree?

- to get an unbiased estimate of a learned model's accuracy, we must use **a set of instances that are held-aside during learning**
- this is called a ***test set***



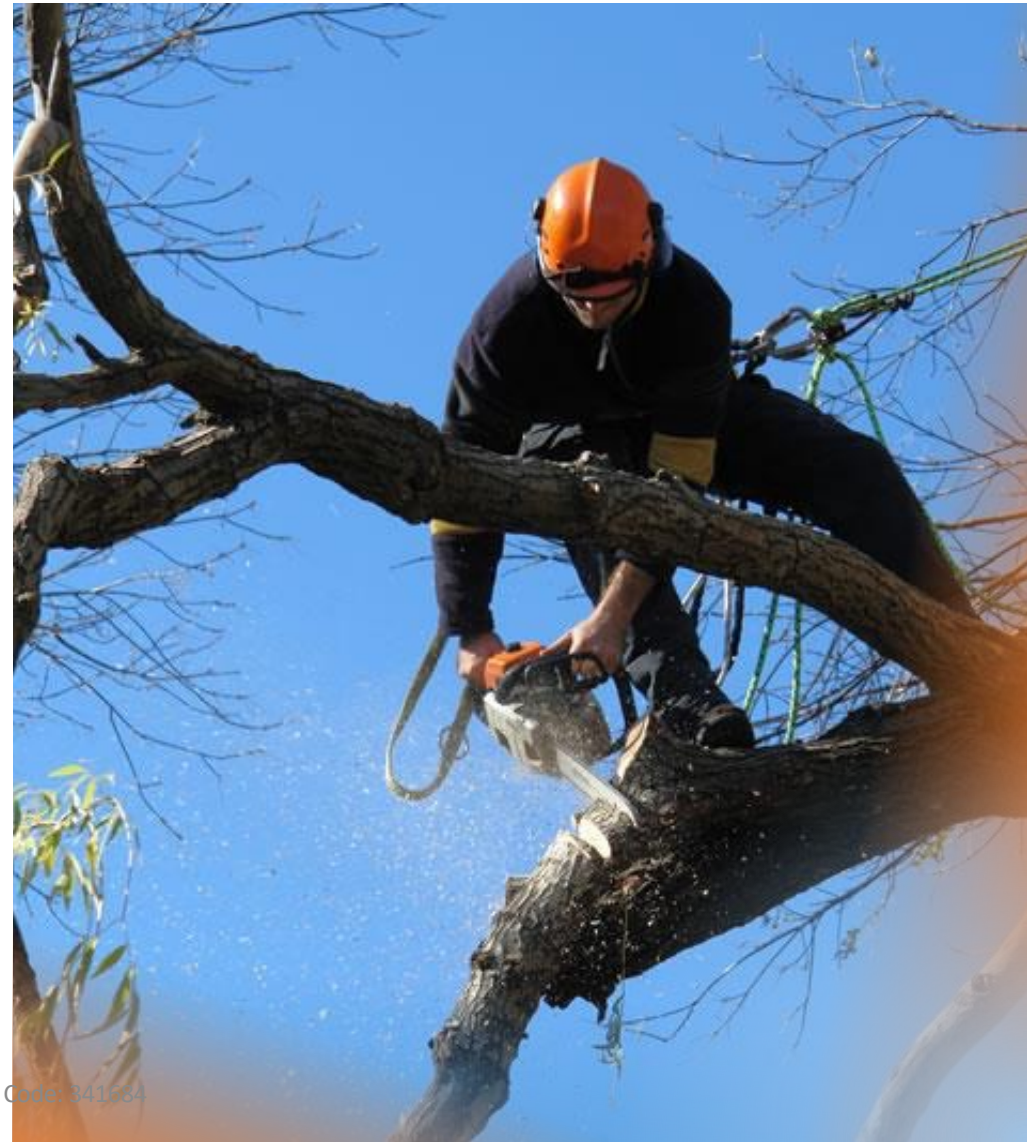


3. Pruning and Validation Dataset

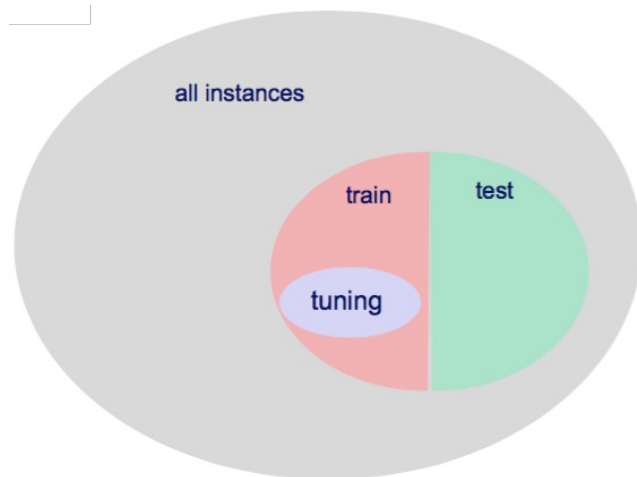
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Stopping criteria

- We should form a leaf when
 - all of the given subset of instances are of the same class
 - we've exhausted all of the candidate splits
- Is there a reason to stop earlier, or to prune back the tree?

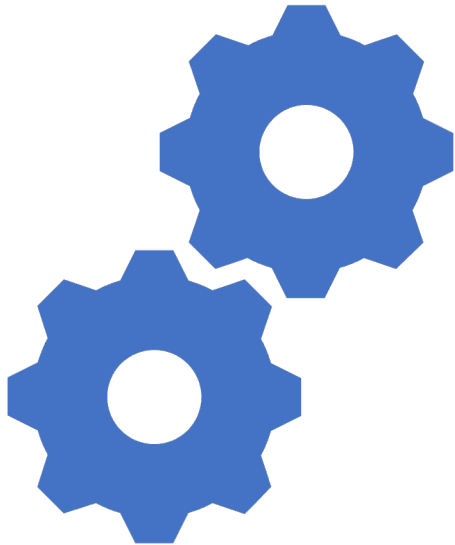


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Pruning in C4.5

- Split given data into training and *validation* (*tuning*) sets
- A *validation set* (a.k.a. *tuning set*) is a subset of the training set that is held aside
 - not used for primary training process (e.g. tree growing)
 - but used to select among models (e.g. trees pruned to varying degrees)



Pruning in C4.5

- Split given data into training and *validation (tuning)* sets
- Grow a complete tree
- do until further pruning is harmful
 - evaluate impact on tuning-set accuracy of pruning each node
 - greedily remove the one that least reduces tuning-set accuracy

Machine Learning isn't just about statistics

- The discovery of adversarial examples (or robustness issues) were started from the below concrete example

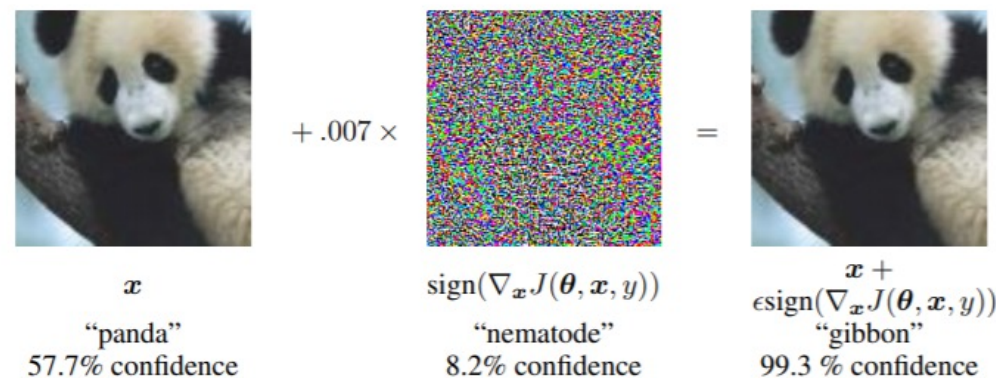


Figure 1: A demonstration of fast adversarial example generation applied to GoogLeNet (Szegedy et al., 2014a) on ImageNet. By adding an imperceptibly small vector whose elements are equal to the sign of the elements of the gradient of the cost function with respect to the input, we can change GoogLeNet's classification of the image. Here our ϵ of .007 corresponds to the magnitude of the smallest bit of an 8 bit image encoding after GoogLeNet's conversion to real numbers.

Explainability aims to identify the reason why a certain classification was made.

- The following was an example from the LIME paper:



(a) Original Image

(b) Explaining *Electric guitar*

(c) Explaining *Acoustic guitar*

(d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception network, highlighting positive pixels. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

More examples

- the importance of not only looking at statistics, why?
- **Worst case performance** is equally, if not more, important!
- Interesting questions **not** covered in this module include e.g., how to estimate the rare event probability?, and how to use the estimation for the improvement of the training to reduce the rare events?

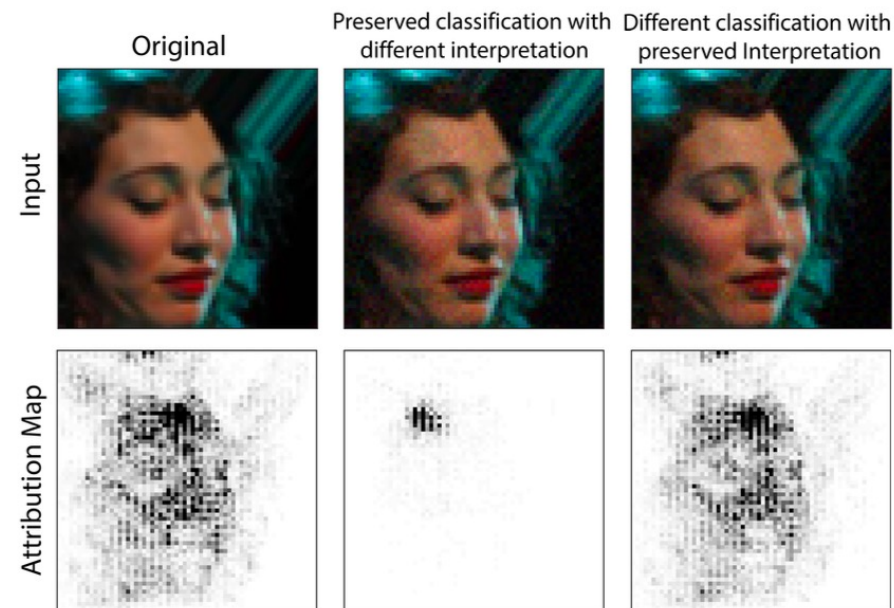


Figure 1: Two types of misinterpretations after perturbation

From our research published in ICCV2023.

Let's take a look at another example on overfitting.

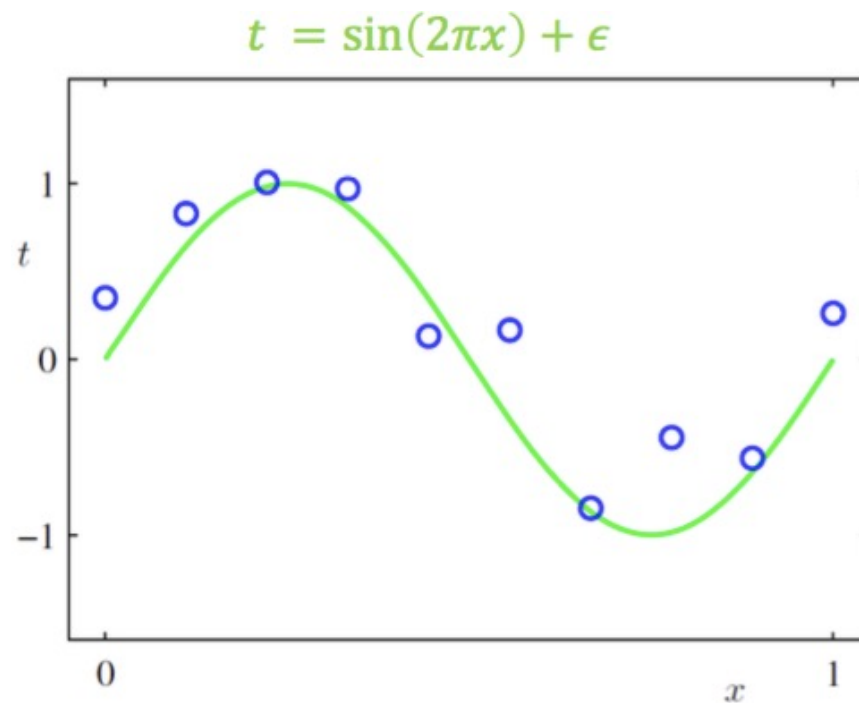
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4. Overfitting

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Example 3: regression using polynomial

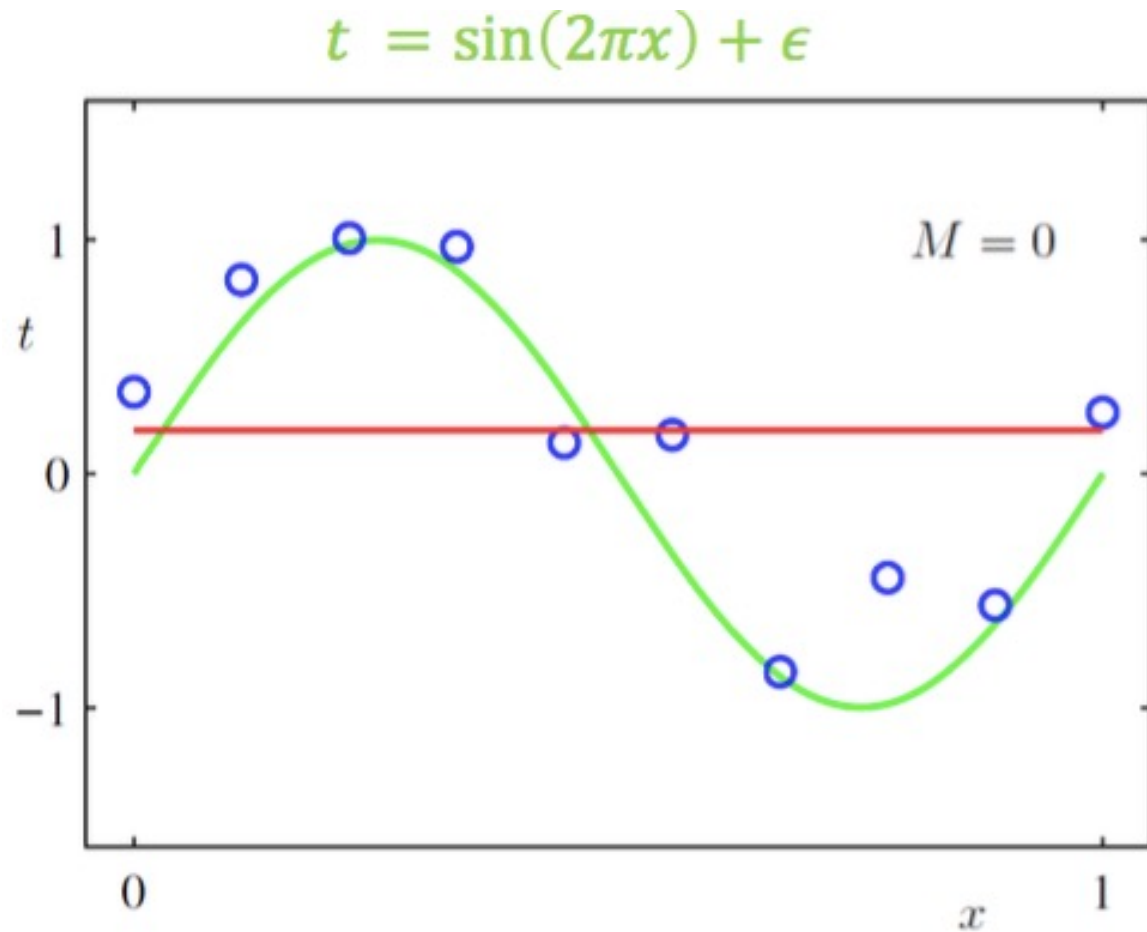


Regression using
polynomial of
degree M

$$y = w_M x^M + w_{M-1} x^{M-1} + \dots + w_1 x + w_0$$

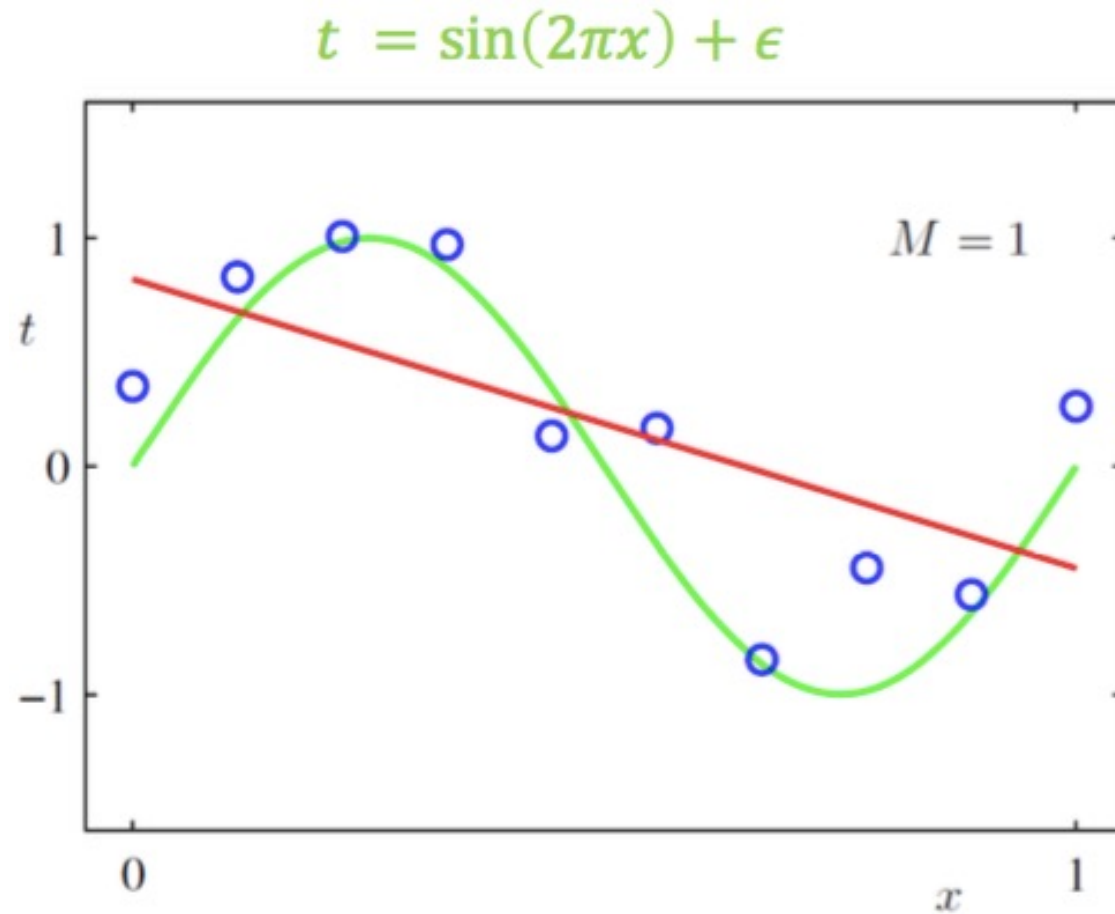
Example 3:
regression using
polynomial

$$y = c$$



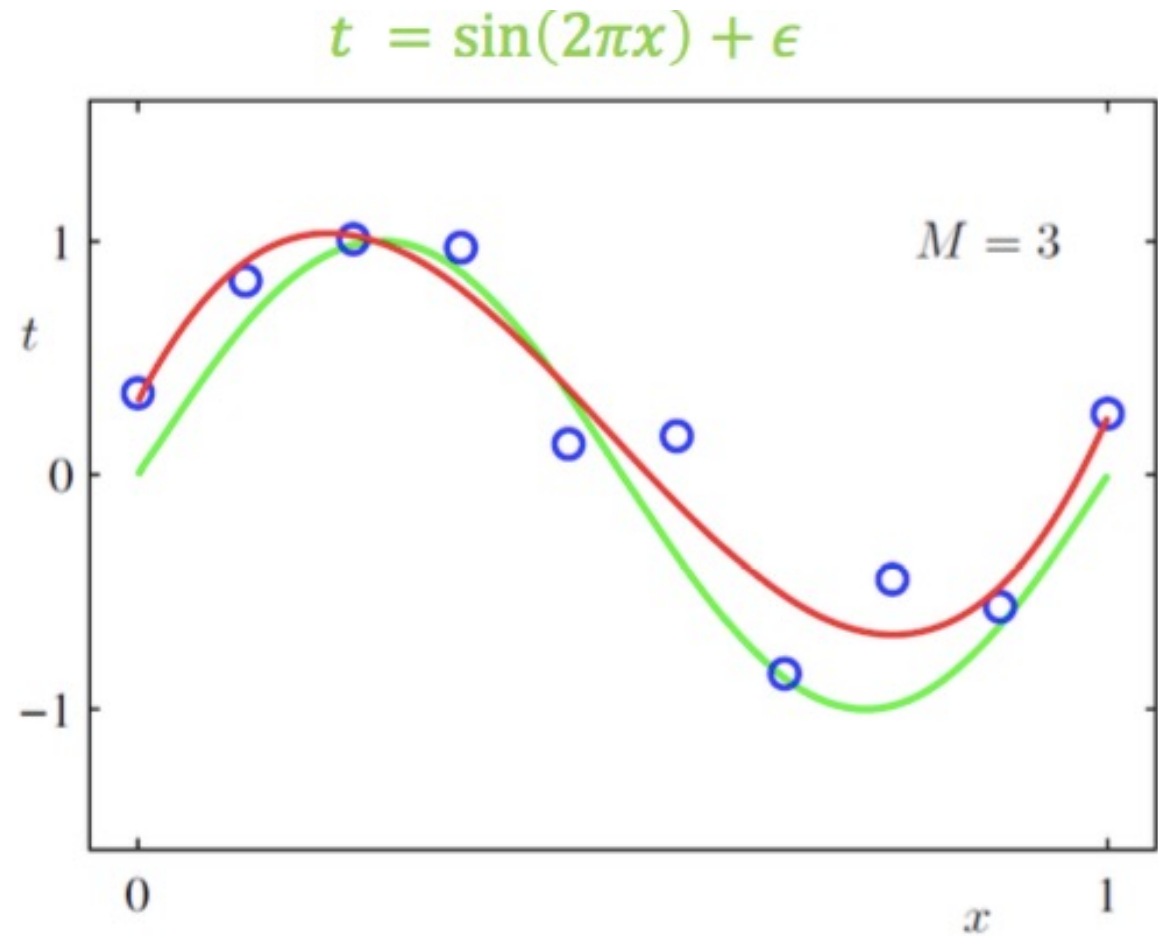
Example 3: regression using polynomial

$$y = w_1x + w_0$$



Example 3:
regression using
polynomial

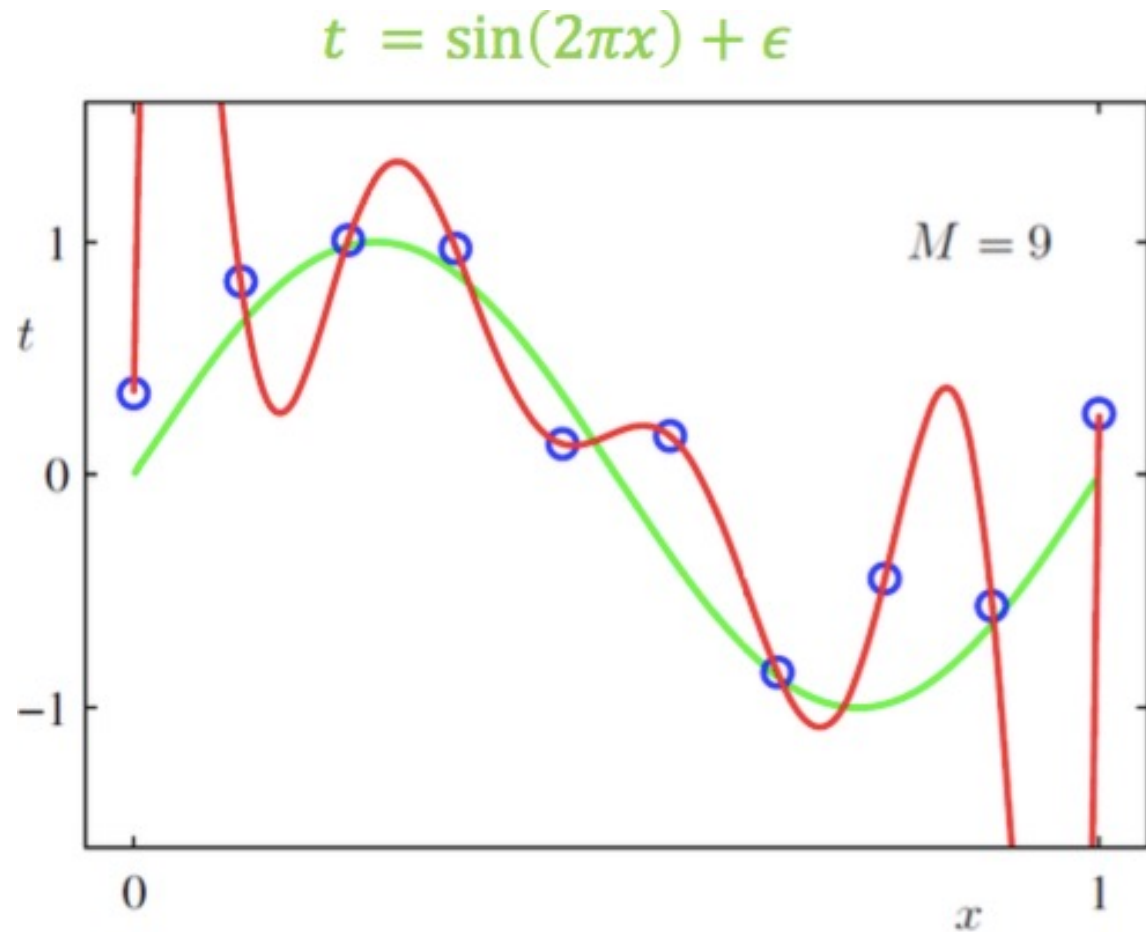
$$y = w_3x^3 + w_2x^2 + w_1x + w_0$$



Example 3: regression using polynomial

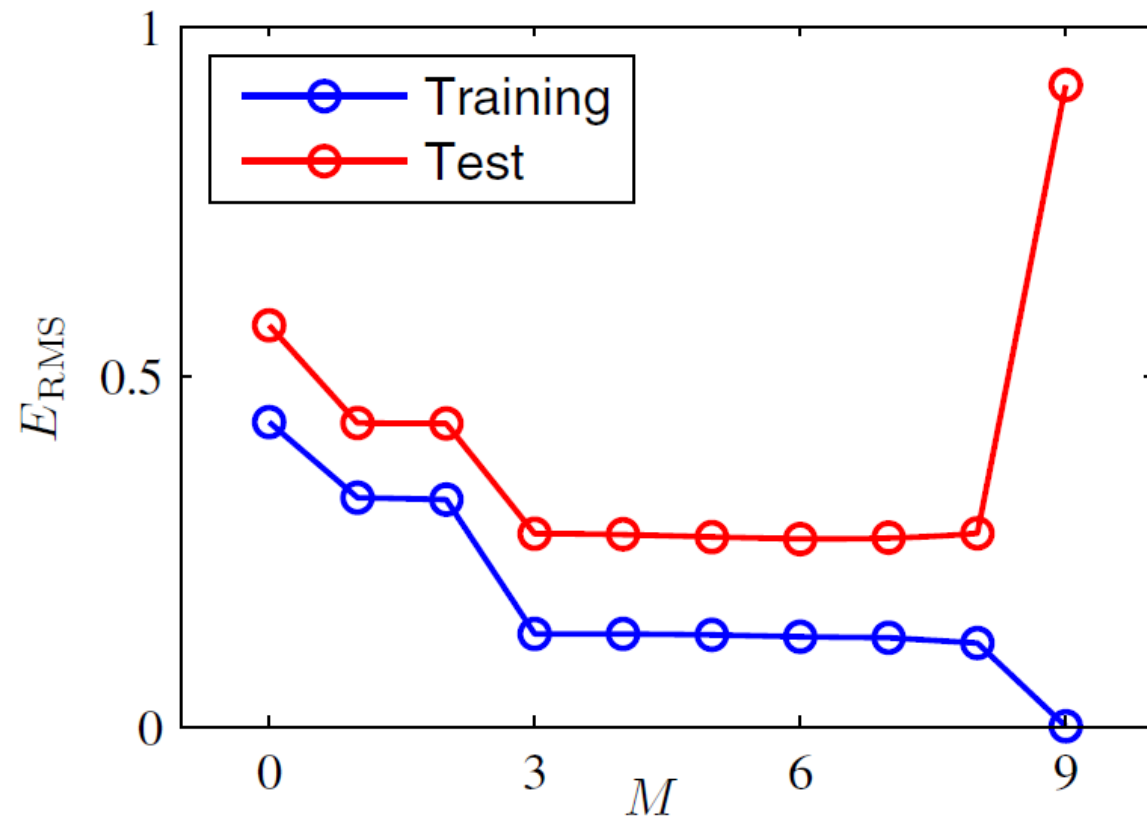
$$y = w_9x^9 + w_8x^8 + \dots + w_1x + w_0$$

Overfits, why?

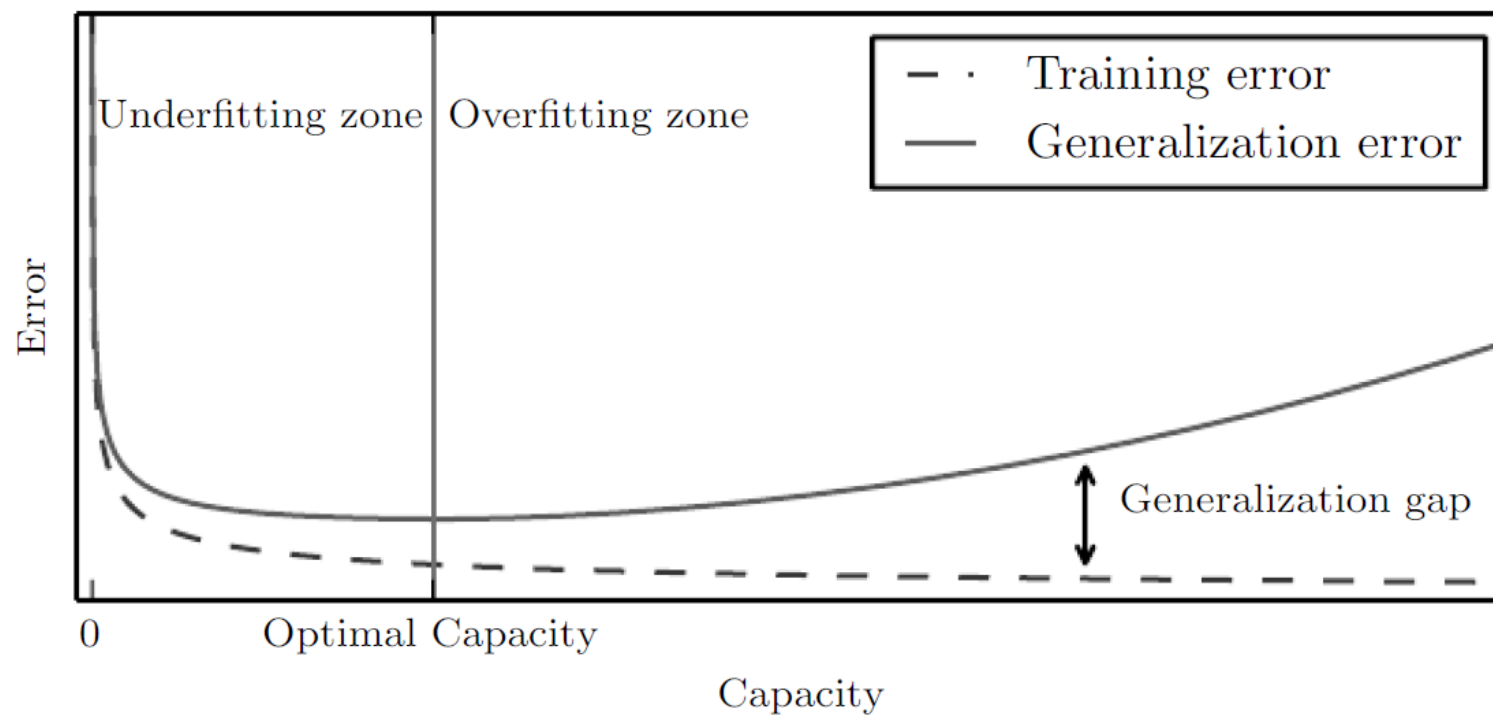


Example: regression using polynomial

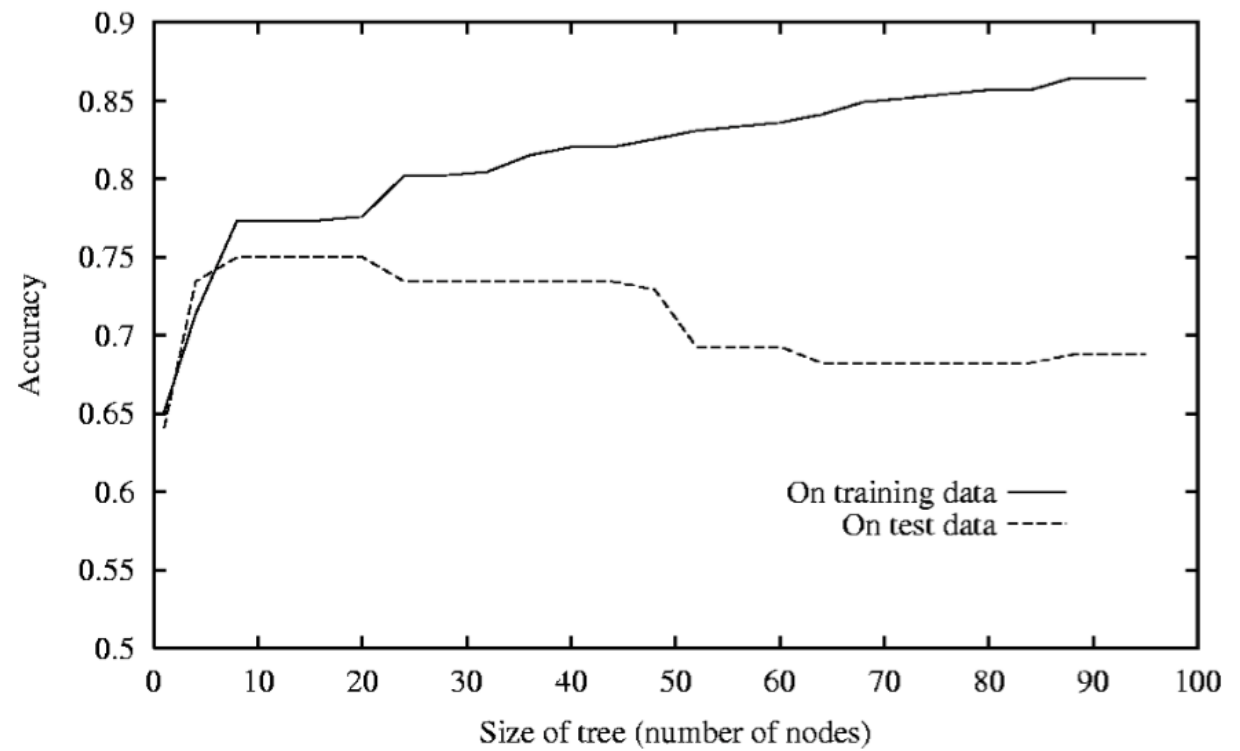
RMS: root mean square,
i.e., the [square root](#) of
the [mean square](#)



General phenomenon



Overfitting in decision trees



Prevent overfitting

Will have examples in decision tree later for these two cases

- Cause: training error and expected error are different

- there may be noise in the training data
- training data is of limited size, resulting in difference from the true distribution
- the larger the hypothesis class, the easier to find a hypothesis that fits the difference between the training data and the true distribution

Small training dataset

Large model

- How to reduce overfitting:
 - cleaner training data help!
 - more training data help!
 - throwing away unnecessary hypotheses helps! (Occam's Razor)

5. Overfitting in Decision Tree



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Overfitting

- Consider error of model M over
 - training data: $Error(D_{training}, M)$
 - entire distribution of data: $Error(D_{true}, M)$
- Model $M \in H$ **overfits** the training data if there is an alternative model $M' \in H$ such that

$$Error(D_{training}, M) < Error(D_{training}, M')$$

Perform better on
training dataset

$$Error(D_{true}, M) > Error(D_{true}, M')$$

Perform worse on
true distribution

Example 1: overfitting with noisy data

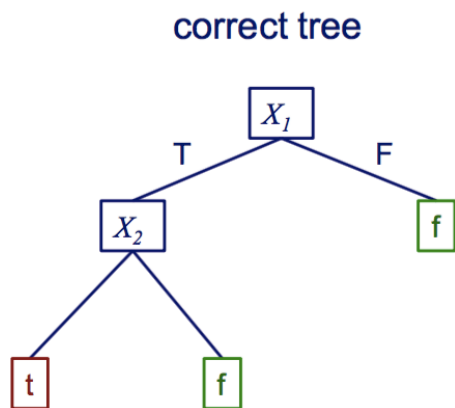
- suppose

- the target concept is $Y = X_1 \wedge X_2$ Ground truth
- there is noise in some feature values
- we're given the following training set

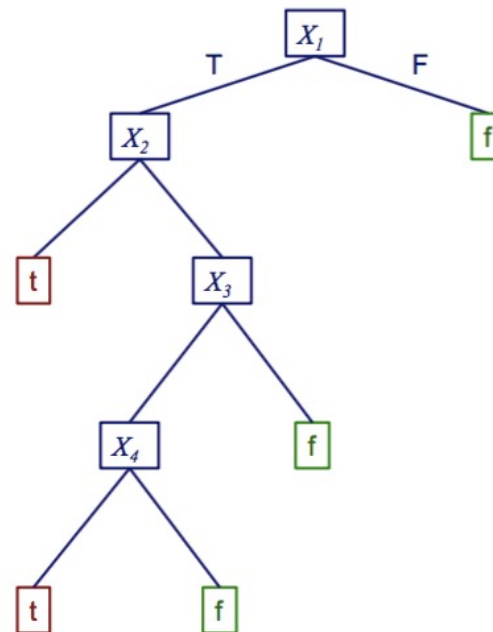
X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	f	f	t	...	t
t	f	t	t	f	...	t
t	f	f	t	f	...	f
t	f	t	f	f	...	f
f	t	t	f	t	...	f

noisy value

Example 1: overfitting with noisy data



tree that fits noisy training data



$$Y = X_1 \wedge X_2$$

A noisy data sample:

$X_1 = t$

$X_2 = f$

$X_3 = t$

$X_4 = t$

$X_5 = f$

$Y = t$

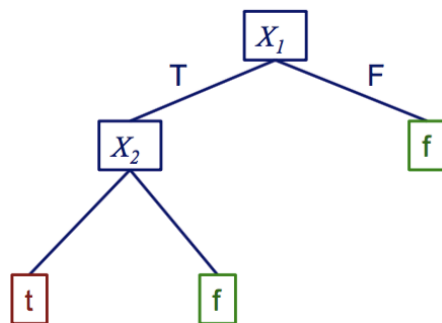
X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	f	f	t	...	t
t	f	t	t	f	...	t
t	f	f	t	f	...	f
t	f	t	f	f	...	f
f	t	t	f	t	...	f

noisy value

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Example 1: overfitting with noisy data

correct tree



What is the accuracy?

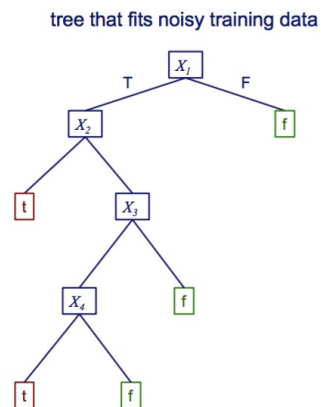
- $\text{Accuracy}(D_{\text{training}}, M) = 5/6$
- $\text{Accuracy}(D_{\text{true}}, M) = 100\%$

$$Y = X_1 \wedge X_2$$

X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	f	f	t	...	t
t	f	t	t	f	...	t
t	f	f	t	f	...	f
t	f	t	f	f	...	f
f	t	t	f	t	...	f

noisy value

Example 1: overfitting with noisy data



$$Y = X_1 \wedge X_2$$

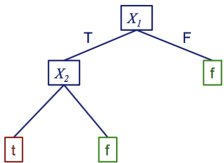
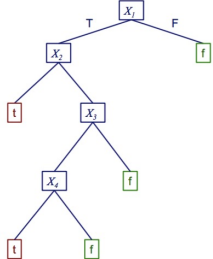
X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	f	f	t	...	t
t	f	t	t	f	...	t
t	f	f	t	f	...	f
t	f	t	f	f	...	f
f	t	t	f	t	...	f

noisy value

What is the accuracy?

- $\text{Accuracy}(D_{\text{training}}, M) = 100\%$
- $\text{Accuracy}(D_{\text{true}}, M) < 100\%$

Example 1: overfitting with noisy data

		Training set accuracy	True accuracy
M_1	correct tree  <pre>graph TD; X1[X1] -- T --> X2[X2]; X1 -- F --> f1[f]; X2 --> t1[t]; X2 --> f2[f]; style t1 fill:#f99; style f2 fill:#9f9;</pre>	5/6	100%
M_2	tree that fits noisy training data  <pre>graph TD; X1[X1] -- T --> X2[X2]; X1 -- F --> f1[f]; X2 --> t1[t]; X2 --> X3[X3]; X3 --> X4[X4]; X3 --> f2[f]; X4 --> t2[t]; X4 --> f3[f]; style t1 fill:#f99; style t2 fill:#f99; style f2 fill:#9f9; style f3 fill:#9f9;</pre>	100%	< 100 %

M_2 is
overfitting!

Example 2: overfitting with noise-free data

- suppose

- the target concept is $Y = X_1 \wedge X_2$
- $P(X_3 = t) = 0.5$ for both classes
- $P(Y = t) = 0.66$

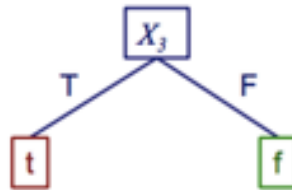
Ground truth

- we're given the following training set

X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	t	f	t	...	t
t	t	t	t	f	...	t
t	f	f	t	f	...	f
f	t	f	f	t	...	f

To demonstrate:
training data is too
few, leading to fake
correlations between
features and labels

Example 2: overfitting with noise-free data



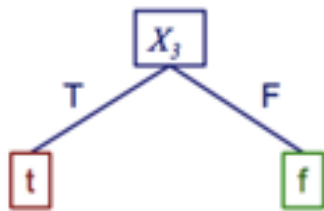
M_1



M_2

X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	t	f	t	...	t
t	t	t	t	f	...	t
t	f	f	t	f	...	f
f	t	f	f	t	...	f

Example 2: overfitting with noise-free data



$$Y = X_1 \wedge X_2$$

$$P(X_3 = t) = 0.5$$

$$P(Y=t) = 0.66$$

What is the accuracy?

- $\text{Accuracy}(D_{\text{training}}, M) = 100\%$
- $\text{Accuracy}(D_{\text{true}}, M) = 50\%$

(X_3, Y) :

$(t, t) = 0.5 \cdot 0.66$ (correct)

$(t, f) = 0.5 \cdot 0.33$ (fail)

$(f, t) = 0.5 \cdot 0.66$ (fail)

$(f, f) = 0.5 \cdot 0.33$ (correct)

So, we have
50% correctness

X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	t	f	t	...	t
t	t	t	t	f	...	t
t	f	f	t	f	...	f
f	t	f	f	t	...	f

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Example 2: overfitting with noise-free data

t

$$Y = X_1 \wedge X_2$$

$$P(X_3 = t) = 0.5$$

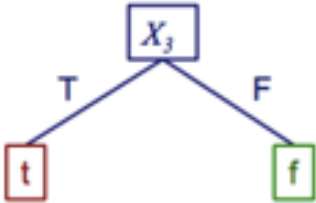
$$P(Y=t) = 0.66$$

What is the accuracy?

- $\text{Accuracy}(D_{\text{training}}, M) = 60\%$
- $\text{Accuracy}(D_{\text{true}}, M) = 66\%$

X_1	X_2	X_3	X_4	X_5	...	Y
t	t	t	t	t	...	t
t	t	t	f	t	...	t
t	t	t	t	f	...	t
t	f	f	t	f	...	f
f	t	f	f	t	...	f

Example 2: overfitting with noise-free data

		Training set accuracy	True accuracy	
M_1		100%	50%	M_1 is overfitting!
M_2		60%	66%	

because the training set is a limited sample, there might be (combinations of) features that are correlated with the target concept by chance

Avoiding overfitting in DT learning

The stopping criterion utilizing validation dataset

- two general strategies to avoid overfitting
 1. *early stopping*: stop if further splitting not justified by a statistical test
 - Quinlan's original approach in ID3
 2. *post-pruning*: grow a large tree, then prune back some nodes
 - more robust to myopia of greedy tree learning